

map, click on **Save** and give an appropriate comment to describe the changes that you made. Subsequent changes in the same session do not require any new comments.

Editing OSM using JOSM

JOSM is a Java application that can be used to edit OSM. It's the most popular program used to edit OpenStreetMap. It's very easy to install and use, and it's really fun too. Download JOSM from <http://josm.openstreetmap.de/josm-tested.jar> (a Windows installer is at <http://josm.openstreetmap.de/download/windows/josm-setup.exe>) to install it via the Ubuntu package manager, and simply *apt-get* it from the repositories.

You may want to upgrade the package later. To update to the latest JOSM version from the package maintainer's version, download the JAR file from the JOSM site, and copy it to `/usr/share/josm` with the file name *josm.jar*.

Running JOSM

Go to the directory in which the JAR file has been downloaded, then run *java -jar josm-tested.jar* or on Ubuntu, run `/usr/bin/josm`.

Using JOSM

Before you do any editing, you could download a base map to start working on. Go to **File->Download From OSM** (**Ctrl+Shift+down arrow key**) and select a small area you want to work on. This area could be your town or city. After that, you would be able to see the features of the map that have already been done. You are now ready to start mapping.

To pan over the map, right-click and drag. You'd create nodes by double-clicking, and paths by clicking along the path (ending in a double-click), and polygons by clicking along the sides of the polygon—ending by clicking again on the start node. You can switch between drawing mode and select mode by pressing '**a**' for drawing mode and '**s**' for select mode.



Potlatch loading on Firefox



Potlatch map edit view

In the select mode, you can click on points, delete them, move them around, or rotate them.

After drawing the nodes, you can add properties to them by clicking on the **preset** menu. To add more attributes, locate the **Properties/Memberships** window (**Alt + Shift + P** to toggle) and click the **Add** button. You can choose from a plethora of available keys and values. When necessary, you can also add custom values. For a road, a key may be *surface type* and a value may be *Gravel* or *Asphalt*.

Become a Ninja

Exploration and discovery is the best way to learn. You can learn a great

deal about JOSM by exploring the various tools and options. Check out the various windows that JOSM has, and find out what each one of them is for. Experiment a lot before you do an actual data commit. This way, you get to become familiar with the basics of OSM and mapping. Once you get comfortable with the interface and the concepts, you will be ready to go ahead and do the advanced stuff on the path to becoming a mapping Ninja (like a black belt for mappers; and in case you're wondering, I flaunt a white belt!)

Once you're done, upload the data to the OSM servers by clicking **File->Upload data** (**Ctrl+Shift+Up arrow**).